Model Counting with Algebraic Decision Diagrams

Vu Phan – COMP 600 (Rice University)

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Joint work with Moshe Vardi and Jeffrey Dudek

Overview: Model Counting

Satisfiability problem (SAT): whether Boolean formula has satisfying assignment

- Complexity: NP-complete [Cook, 1971]

Model counting problem (#SAT): number of satisfying assignments of Boolean formula

- Complexity: #P-complete [Valiant, 1979]
- Applications:
  - Hardware verification [Naveh et al., 2007]
  - Bayesian inference [Sang et al., 2005]
    - Medical diagnosis [Shwe et al., 1991]

Contents

1. Model Counting and Bayesian Inference
2. Algebraic Decision Diagrams for Model Counting
3. Empirical Evaluation

Progress

1. Model Counting and Bayesian Inference
2. Algebraic Decision Diagrams for Model Counting
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Boolean Logic Syntax

Boolean formula (in Conjunctive Normal Form):

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

- Boolean variables: x_1, x_2, x_3
- Positive literals: x_1, x_2
- Negative literals: $\neg x_2, \neg x_3$

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- Positive literals: x_1, x_2
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- Disjunctions (clauses): $x_1, \neg x_2, (x_1 \vee \neg x_2), (x_1 \vee x_2 \vee \neg x_3)$

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- Boolean variables: x_1, x_2, x_3
- Positive literals: x_1, x_2
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- Disjunctions (clauses): $x_1, \neg x_2, (x_1 \vee \neg x_2), (x_1 \vee x_2 \vee \neg x_3)$
- Conjunction (formula): $x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$

Model Counting Problem [Valiant, 1979]

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$
$$\varphi : \mathbb{B}^3 \to \mathbb{B}$$

x_1	x_2	x_3	$\varphi(x_1, x_2, x_3)$	Satisfying assignment?
T	T	T	F	No
T	T	F	F	No
T	F	T	**T**	**Yes**, model $M_1 = \{x_1, \neg x_2, x_3\}$
T	F	F	**T**	**Yes**, model $M_2 = \{x_1, \neg x_2, \neg x_3\}$
F	T	T	F	No
F	T	F	F	No
F	F	T	F	No
F	F	F	F	No

Model count:

$$\#\varphi = \mathbf{2}$$

Weighted Model Counting Problem

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

Literal weights	
$W(x_1) = 1.0$	$W(\neg x_1) = 1.0$
$W(x_2) = 0.2$	$W(\neg x_2) = 0.8$
$W(x_3) = 0.3$	$W(\neg x_3) = 0.7$

Weighted Model Counting Problem

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

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Model weights:

$$W(M_1) = W(\{x_1, \neg x_2, x_3\}) = W(x_1) \cdot W(\neg x_2) \cdot W(x_3) = 1.0 \cdot 0.8 \cdot 0.3 = 0.24$$
$$W(M_2) = W(\{x_1, \neg x_2, \neg x_3\}) = W(x_1) \cdot W(\neg x_2) \cdot W(\neg x_3) = 1.0 \cdot 0.8 \cdot 0.7 = 0.56$$

Weighted Model Counting Problem

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Model weights:

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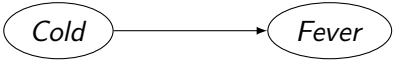
$$W(M_2) = W(\{x_1, \neg x_2, \neg x_3\}) \quad = W(x_1) \cdot W(\neg x_2) \cdot W(\neg x_3) \quad = 1.0 \cdot 0.8 \cdot 0.7 = 0.56$$

Weighted model count:

$$W(\varphi) = W(M_1) + W(M_2) = 0.24 + 0.56 = 0.8$$

Bayesian Network [Pearl, 1985]

Bayesian network:



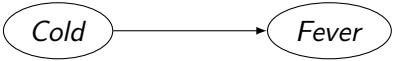
$$P(Cold) = 0.4$$

$$P(Fever|Cold) = 0.3$$
$$P(Fever|\neg Cold) = 0.1$$

Query:

$$P(Cold|Fever) = \ ?$$

Bayesian Network: Reduction to Model Counting [Sang et al., 2005]



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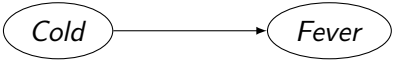

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$W(x_2) = 0.3$	$W(\neg x_2) = 0.7$
$W(x_3) = 0.1$	$W(\neg x_3) = 0.9$
$W(x_4) = 1.0$	$W(\neg x_4) = 1.0$

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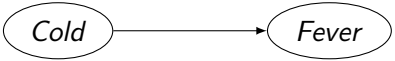

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$W(x_4) = 1.0$	$W(\neg x_4) = 1.0$

$$\varphi = (\neg x_1 \vee \neg x_2 \vee x_4) \wedge (\neg x_1 \vee x_2 \vee \neg x_4) \wedge (x_1 \vee \neg x_3 \vee x_4) \wedge (x_1 \vee x_3 \vee \neg x_4) \wedge x_4$$

$$\gamma = \varphi \wedge x_1$$

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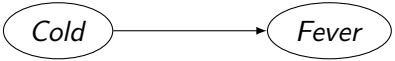

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$$\varphi = (\neg x_1 \vee \neg x_2 \vee x_4) \wedge (\neg x_1 \vee x_2 \vee \neg x_4) \wedge (x_1 \vee \neg x_3 \vee x_4) \wedge (x_1 \vee x_3 \vee \neg x_4) \wedge x_4$$

$$\gamma = \varphi \wedge x_1$$

Computing answer to query:

$$P(Cold|Fever) = \frac{P(Fever, Cold)}{P(Fever)} = \frac{W(\gamma)}{W(\varphi)} = \frac{0.12}{0.18} = 0.67$$

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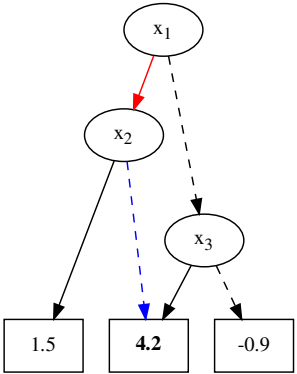
Algebraic Decision Diagram (ADD) [Bahar et al., 1997]

Function $f : \mathbb{B}^3 \to \mathbb{R}$

Exhaustive enumeration

x_1	x_2	x_3	$f(x_1, x_2, x_3)$
T	T	T	1.5
T	T	F	1.5
T	F	T	**4.2**
T	F	F	**4.2**
F	T	T	4.2
F	T	F	-0.9
F	F	T	4.2
F	F	F	-0.9

Algebraic Decision Diagram (ADD)



Variable Eliminations and Function Weight

Arbitrary function:

$$f : \mathbb{B}^2 \to \mathbb{R}$$

Arbitrary real-valued literal weights:

$W(x_1)$	$W(\neg x_1)$
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Eliminating variable x_2 from $f(x_1, x_2)$:

$$g = \sum_{x_2}^{W} f \qquad (g : \mathbb{B}^1 \to \mathbb{R})$$

$$g(x_1) = f(x_1, \mathsf{T}) \cdot W(x_2) + f(x_1, \mathsf{F}) \cdot W(\neg x_2)$$

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Eliminating variable x_1 from $g(x_1)$:

$$h = \sum_{x_1}^{W} g \qquad (h : \mathbb{B}^0 \to \mathbb{R})$$

$$h() = g(\mathsf{T}) \cdot W(x_1) + g(\mathsf{F}) \cdot W(\neg x_1)$$

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Function weight:

$$W(f) = h()$$

Algebraic Decision Diagrams (ADDs) for Model Counting

Boolean formula $\varphi : \mathbb{B}^3 \to \mathbb{B}$

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Widen codomain of φ to create $f : \mathbb{B}^3 \to \mathbb{R}$

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Compute function weight:

$$W(f) = \left(\sum_{x_1}^{W} \sum_{x_2}^{W} \sum_{x_3}^{W} f \right) ()$$

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(Represent functions with ADDs)

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Widen codomain of φ to create $f : \mathbb{B}^3 \to \mathbb{R}$

Compute function weight:

$$W(f) = \left(\sum_{x_1}^{W} \sum_{x_2}^{W} \sum_{x_3}^{W} f \right) ()$$

(Represent functions with ADDs)

Computing answer to query:

$$W(\varphi) = W(f)$$

Representations: Monolithic versus Factored

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

Naive approach: using monolithic representation

- Builds a large ADD for the whole formula φ

$$D(x_1, x_2, x_3) = \texttt{ADD}\,(\varphi)$$

- Scales poorly for complex formulas
  - $|ADD_nodes| = O(\exp(|formula_variables|))$

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  - $|ADD_nodes| = O(\exp(|formula_variables|))$

Our approach: exploiting factored representation

- Builds a small ADD for each clause
- Combines ADDs iteratively
  - Eliminates variables as soon as possible to reduce sizes of ADDs

Exploiting Factored Representation in Conjunctive Normal Form

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

Exploiting Factored Representation in Conjunctive Normal Form

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

Build and combine ADDs:

$$D_1(x_1, x_2) = \sum_{x_3} \texttt{ADD}\,(x_1 \vee x_2 \vee \neg x_3)$$

$$D_2(x_1) = \sum_{x_2} (D_1(x_1, x_2) \cdot \texttt{ADD}\,(\neg x_2) \cdot \texttt{ADD}\,(x_1 \vee \neg x_2))$$

$$D_3() = \sum_{x_1} (D_2(x_1) \cdot \texttt{ADD}\,(x_1))$$

Exploiting Factored Representation in Conjunctive Normal Form

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$$D_3() = \sum_{x_1} (D_2(x_1) \cdot \texttt{ADD}\,(x_1))$$

Heuristics:

- Grouping clauses into clusters: $\kappa_1 = \langle x_1 \rangle$, $\kappa_2 = \langle \neg x_2, x_1 \vee \neg x_2 \rangle$, $\kappa_3 = \langle x_1 \vee x_2 \vee \neg x_3 \rangle$

Exploiting Factored Representation in Conjunctive Normal Form

$$\varphi = x_1 \wedge \neg x_2 \wedge (x_1 \vee \neg x_2) \wedge (x_1 \vee x_2 \vee \neg x_3)$$

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- Grouping clauses into clusters: $\kappa_1 = \langle x_1 \rangle$, $\kappa_2 = \langle \neg x_2, x_1 \vee \neg x_2 \rangle$, $\kappa_3 = \langle x_1 \vee x_2 \vee \neg x_3 \rangle$
- Combining clusters: $\langle \langle \kappa_3, \kappa_2 \rangle, \kappa_1 \rangle$

Contributions

- Algorithm for weighted model counting using Algebraic Decision Diagrams (ADDs)
  - Exploits factored representation in Conjunctive Normal Form: $\varphi = c_1 \wedge c_2 \wedge \ldots$
    - Builds small clause ADDs then combines them
    - Eliminates variables early
  - Utilizes various heuristics
- Tool: Algebraic Decision Diagram Model Counter (`ADDMC`)

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Comparing Weighted Model Counters

Setup:

- Rice NOTS Linux cluster
- Timeout: 1000-second

Table: 1091 standard weighted model counting benchmarks

Weighted model counter	Benchmarks solved	Percentage solved
`Cachet` [Sang et al., 2004]	776	71%
`miniC2D` [Oztok and Darwiche, 2015]	913	84%
`ADDMC` (our tool)	1085	99%

Comparing Weighted Model Counters

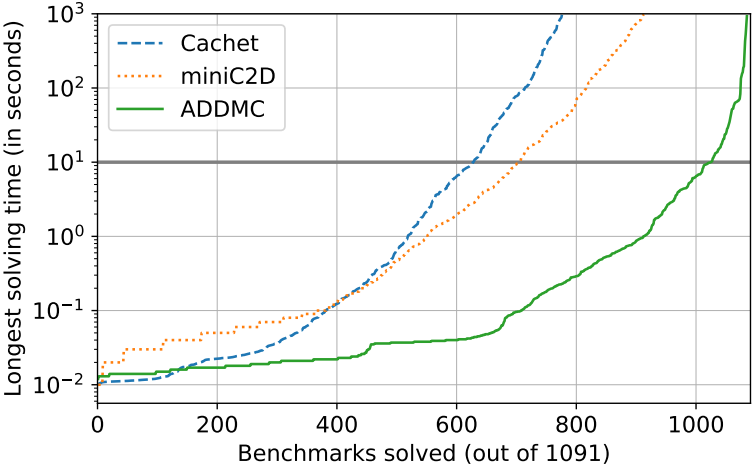

Figure: Cactus plot (rightmost is best)

Conclusion

- Problem: weighted model counting
  - Theoretical hardness: #P-complete
- Our approach: using ADDs, exploiting factored representation
  - Practical efficiency: outperforming other weighted model counters
- Future work:
  - Arbitrary-precision weighted model counting
  - Multi-core computing

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